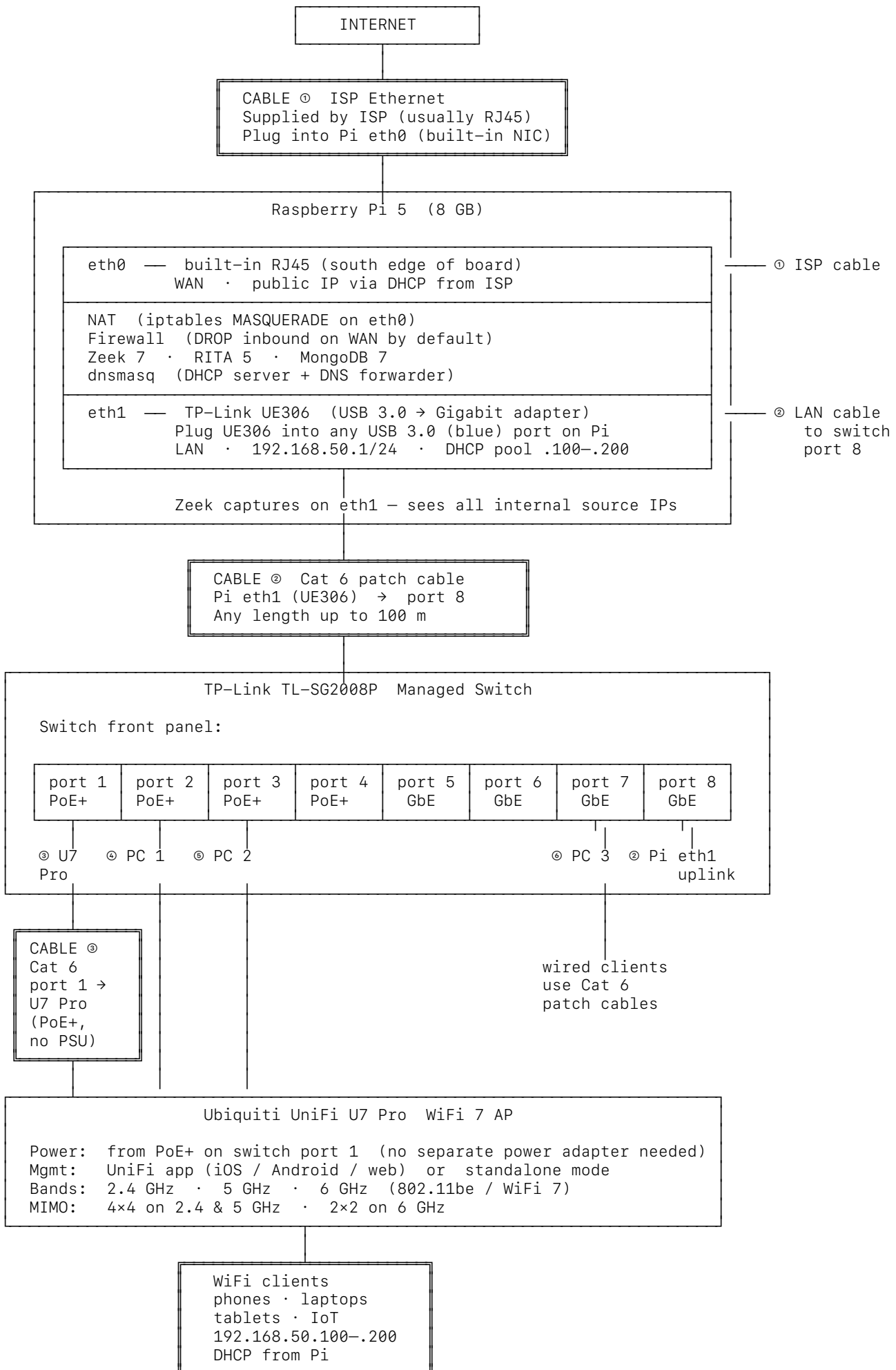


BeaconButty – Physical Network Diagram & Cabling Guide



SWITCH PORT ASSIGNMENTS

Port 1	PoE+	Ubiquiti U7 Pro WiFi 7 AP (cable ①)
Port 2	PoE+	Wired client or second AP if needed (cable ②)
Port 3	PoE+	Wired client (cable ③)
Port 4	PoE+	Wired client (cable ④ – or spare PoE for future device)
Port 5	GbE	Wired client
Port 6	GbE	Wired client
Port 7	GbE	Wired client
Port 8	GbE	Pi eth1 uplink (cable ⑤)

CABLING SEQUENCE

- ① ISP cable → Pi eth0 (built-in RJ45, south edge of board)
Do this last – inserting it first means the Pi starts NAT-ing before the LAN is wired up.
- ② Cat 6 → Pi eth1 (UE306 dongle in a blue USB 3.0 port on Pi)
other end → switch port 8 (any non-PoE port works)
Use a blue USB 3.0 port – not a black USB 2.0 port, which caps throughput at 480 Mbps.
- ③ Cat 6 → switch port 1 (PoE+)
other end → U7 Pro LAN/PoE port
No power adapter needed on the AP – PoE feeds it.
- ④⑤⑥ Cat 6 → switch ports 2–7
other end → wired PCs, NAS, printers, etc.
Ports 2–4 also deliver PoE+ if a device needs it.

FIRST-BOOT CHECKLIST

- ☐ Flash Raspberry Pi OS 64-bit (Bookworm) to USB SSD, boot Pi from it
- ☐ SSH to Pi on its initial IP and run: `sudo ./setup.sh`
- ☐ Then run: `sudo ./scripts/07_router_mode.sh` – Pi reboots into router mode
- ☐ Connect a PC to switch port 2, confirm DHCP address is 192.168.50.1xx
- ☐ SSH to 192.168.50.1 and verify:
 - `zeekctl status`
 - `systemctl status dnsmasq`
 - `systemctl status wan-watchdog.timer`
 - `systemctl status rita-analyze.timer`
- ☐ Plug in ISP cable (cable ①) – internet goes live
- ☐ Configure U7 Pro SSID via UniFi app (connect phone to AP, open app)
- ☐ Next morning: `/usr/local/bin/beacon-report.sh` – first overnight analysis

HARDWARE BILL OF MATERIALS

Raspberry Pi 5 8 GB	~£ 80
Pi official USB-C power supply	~£ 12
USB SSD (e.g. Samsung T7, 500 GB)	~£ 55
TP-Link UE306 USB 3.0 Gigabit adapter	~£ 12
TP-Link TL-SG2008P 8-port managed switch	~£ 55
Ubiquiti UniFi U7 Pro WiFi 7 access point	~£200
Cat 6 patch cables (×4, assorted lengths)	~£ 15
Total	~£429